

Hospital Emergency Room Dashboard

Power BI Project Report

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1. Executive Summary

The Hospital Emergency Room Dashboard is an interactive Power BI solution developed to monitor and analyze Emergency Room (ER) performance. The dashboard provides healthcare stakeholders with real-time insights into patient flow, operational efficiency, service quality, and demographic trends.

The project transforms raw patient records into meaningful visualizations that support faster decision-making, improve resource planning, reduce patient wait times, and enhance the overall patient experience.

The solution consists of four interactive dashboards:

- Monthly View
- Consolidated View
- Patient Details
- Key Takeaways

2. Project Overview

Healthcare organizations handle thousands of emergency cases every month. Without proper reporting, it becomes difficult to identify operational bottlenecks, monitor patient satisfaction, or allocate resources effectively.

This Power BI dashboard addresses these challenges by providing interactive reports that enable hospital administrators to:

- Monitor patient admissions.
- Track emergency room performance.
- Analyze patient demographics.
- Measure operational efficiency.
- Improve patient satisfaction.
- Support strategic decision-making.

3. Business Problem

Hospital management required a centralized reporting solution capable of:

- Tracking Emergency Room patient visits.
- Monitoring average patient wait time.
- Measuring patient satisfaction.
- Analyzing referral patterns.
- Understanding demographic distribution.
- Identifying peak operational hours.
- Supporting better staffing and resource allocation.

The objective was to replace manual reporting with a dynamic Power BI dashboard.

4. Business Requirements

The dashboard was designed to answer the following business questions.

Key Performance Indicators (KPIs)

1. Number of Patients

- Total ER patient visits.
- Daily patient trend using sparkline.
- Monthly patient comparison.

2. Average Wait Time

- Average waiting time before treatment.
- Daily trend analysis.
- Identification of peak waiting periods.

3. Patient Satisfaction Score

- Average patient satisfaction rating.
- Daily satisfaction trend.
- Relationship between waiting time and satisfaction.

4. Number of Patients Referred

- Total referrals by department.
- Daily referral trend.
- Department-wise referral distribution.

5. Dataset Description

The dataset contains patient-level Emergency Room information.

Field	Description
Patient ID	Unique patient identifier
Admission Date	Date of ER visit
First Initial	Patient first initial
Last Name	Patient surname
Gender	Male, Female, Other
Age	Patient age
Race	Patient ethnicity
Department Referral	Department referred to
Admission Flag	Admitted or Not Admitted
Satisfaction Score	Patient feedback score
Wait Time	Waiting time before consultation
Case Manager	Assigned care coordinator

6. Project Methodology

The project was completed using the following Business Intelligence workflow:

1. Requirement Gathering
2. Data Understanding
3. Data Connection (Power BI)
4. Data Cleaning (Power Query)
5. Data Modeling (Star Schema)
6. Data Processing
7. DAX Measure Creation
8. Dashboard Design & Development
9. Insight Generation

7. Data Modeling

The data model follows a star schema.

Fact Table:

- Hospital ER Records

Dimension Tables:

- Date
- Department
- Gender
- Race
- Admission Status

Benefits:

- Faster report performance
- Better scalability
- Simplified DAX calculations
- Improved filtering

8. DAX Measures

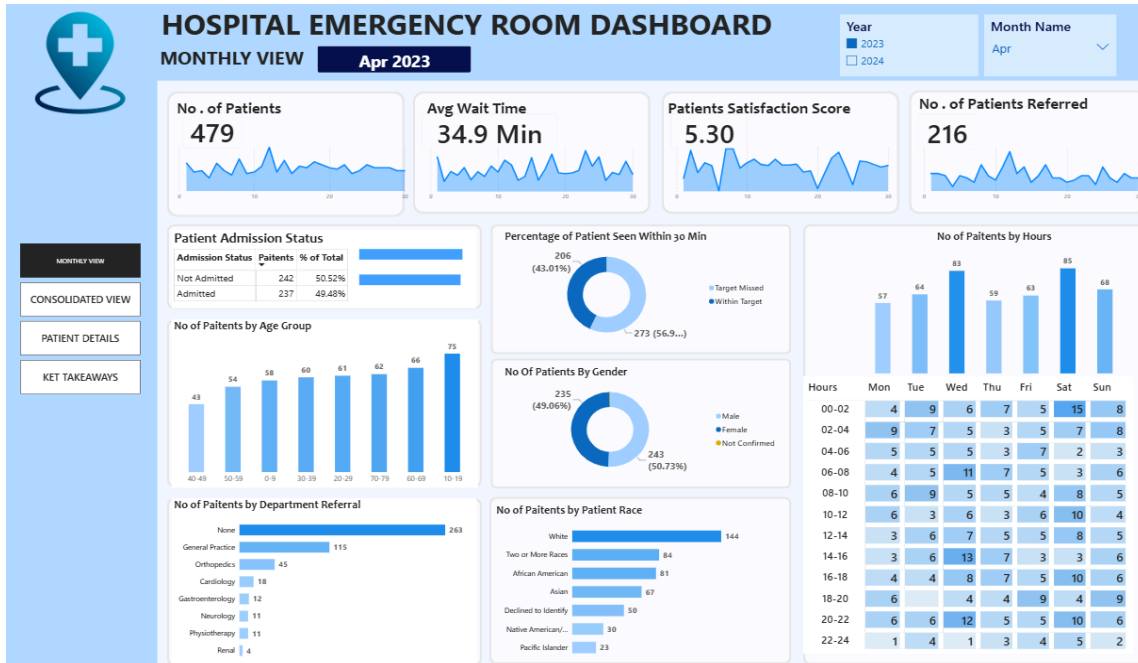
Examples of measures created:

- Total Patients
- Total Admitted Patients
- Total Non-Admitted Patients
- Average Wait Time
- Average Satisfaction Score
- Referral Count
- Admission Rate
- Patients Seen Within 30 Minutes
- Timeliness Percentage

These measures allow dynamic calculations based on selected filters and date ranges.

9. Dashboard Overview

Dashboard 1: Monthly View



Objective

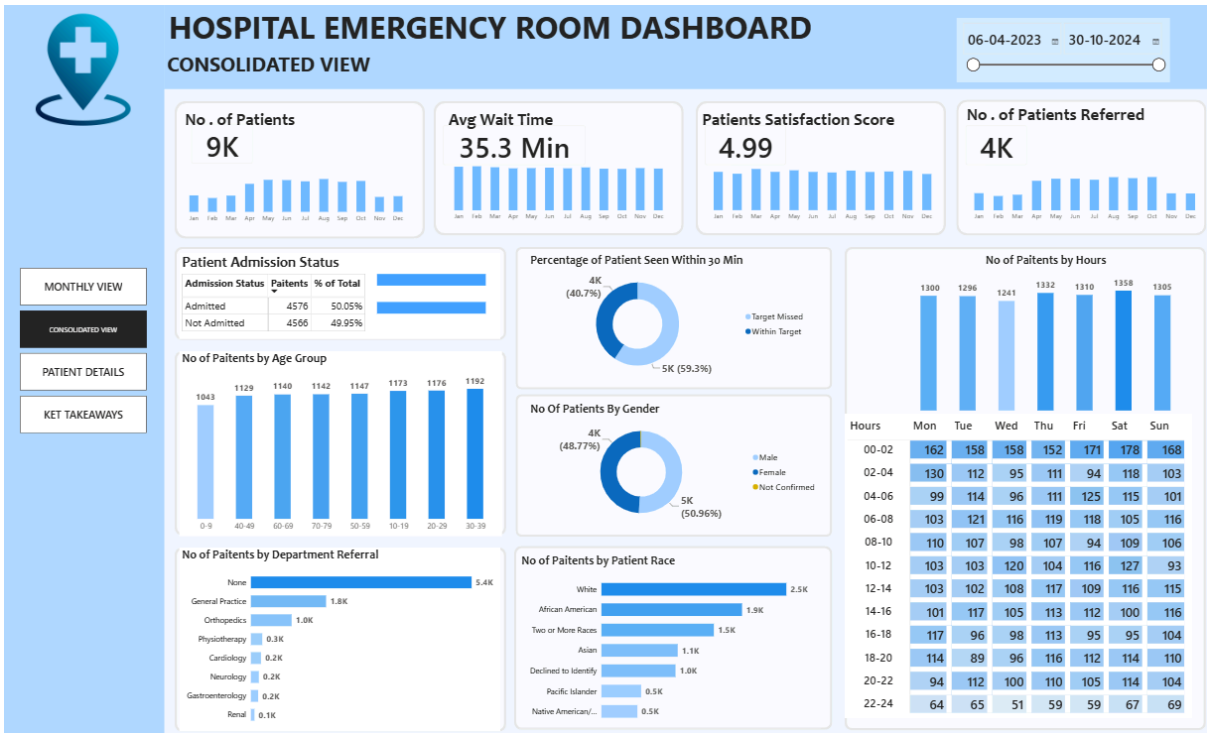
Monitor the Emergency Room's monthly performance and identify trends in patient visits and hospital operations.

Key KPIs

- Total Patients
- Average Wait Time
- Average Satisfaction Score
- Total Department Referrals

The **Monthly View Dashboard** provides a month-wise analysis of Emergency Room performance. It enables users to monitor key performance indicators such as total patient visits, average wait time, patient satisfaction, and department referrals. The dashboard highlights monthly trends in admissions, patient demographics, referral patterns, and peak operating hours, helping hospital management evaluate performance and identify areas that require operational improvements.

Dashboard 2: Consolidated View



Objective

Provide an overall summary of Emergency Room performance for a selected date range.

Key KPIs

- Total Patients
- Average Wait Time
- Average Satisfaction Score
- Total Referrals

The **Consolidated View Dashboard** presents an overall summary of Emergency Room operations for a user-selected date range. It combines all major KPIs and analytical insights into a single interactive report, allowing stakeholders to compare performance across different periods. This dashboard helps management monitor patient flow, analyze demographic trends, evaluate service quality, and make informed decisions based on comprehensive hospital data.

Dashboard 3: Patient Details



HOSPITAL EMERGENCY ROOM DASHBOARD

PATIENT DETAILS

01-04-2023

=

30-10-2024

=

MONTHLY VIEW

CONSOLIDATED VIEW

PATIENT DETAILS

KET TAKEAWAYS

Patient Id	Patient Name	Patient Gender	Patient Age	Patient Admin Date	Patient Race	Patient Waittime	Department Referral	Admission Status
100-04-3993	M St Ange	Female	29	04 April 2023	White	16	None	Not Admitted
100-21-9648	W Marran	Female	39	17 January 2024	Pacific Islander	22	None	Admitted
100-66-8222	F Mullane	Female	65	23 December 2023	Asian	17	General Practice	Not Admitted
100-72-5705	N Dudny	Female	60	19 June 2024	African American	45	None	Admitted
100-74-3943	M Hallard	Female	3	17 September 2023	White	14	None	Admitted
100-74-5636	A Warwicker	Female	47	13 August 2024	Declined to Identify	25	None	Not Admitted
100-79-0109	P Ulyatt	Female	19	27 February 2024	Pacific Islander	36	Neurology	Admitted
100-84-7203	K Ybarra	Female	37	13 June 2023	White	55	None	Not Admitted
101-08-8798	L Willeson	Female	72	31 July 2024	Two or More Races	48	None	Admitted
101-13-4808	V Gowdridge	Female	30	25 April 2023	Declined to Identify	30	None	Admitted
101-35-3930	W Andreotti	Female	30	24 January 2024	White	25	None	Admitted
101-50-7359	D Sabey	Female	50	13 October 2023	White	56	None	Admitted
101-56-3319	M Kilduff	Female	44	08 October 2023	Asian	55	None	Admitted
101-60-5189	G Georger	Female	45	04 August 2024	White	43	None	Not Admitted
101-66-4553	R Odd	Female	14	30 March 2024	Two or More Races	57	General Practice	Admitted
101-85-4810	R McBoyle	Female	74	07 August 2023	White	55	None	Admitted
102-20-6596	V Asson	Female	69	11 June 2024	African American	33	General Practice	Not Admitted
102-21-5953	O Annett	Female	49	14 August 2024	White	11	Orthopedics	Not Admitted
102-54-1653	J Mowson	Female	50	27 April 2024	African American	51	Orthopedics	Not Admitted
102-60-4609	Y Rutt	Female	52	17 January 2024	Declined to Identify	60	General Practice	Not Admitted
102-93-0960	F Lyddiard	Female	47	28 April 2024	Declined to Identify	35	General Practice	Not Admitted
103-16-8002	R Avon	Female	27	14 October 2024	Two or More Races	14	None	Admitted
103-66-1502	I MacBean	Female	20	03 June 2023	White	20	General Practice	Admitted
103-82-8447	M Baldwin	Female	8	23 October 2024	Pacific Islander	52	None	Admitted
103-87-1157	Y Pashby	Female	51	15 September 2023	African American	46	None	Admitted
103-89-7991	U Cranmore	Female	74	11 April 2024	Pacific Islander	34	Physiotherapy	Not Admitted

Objective

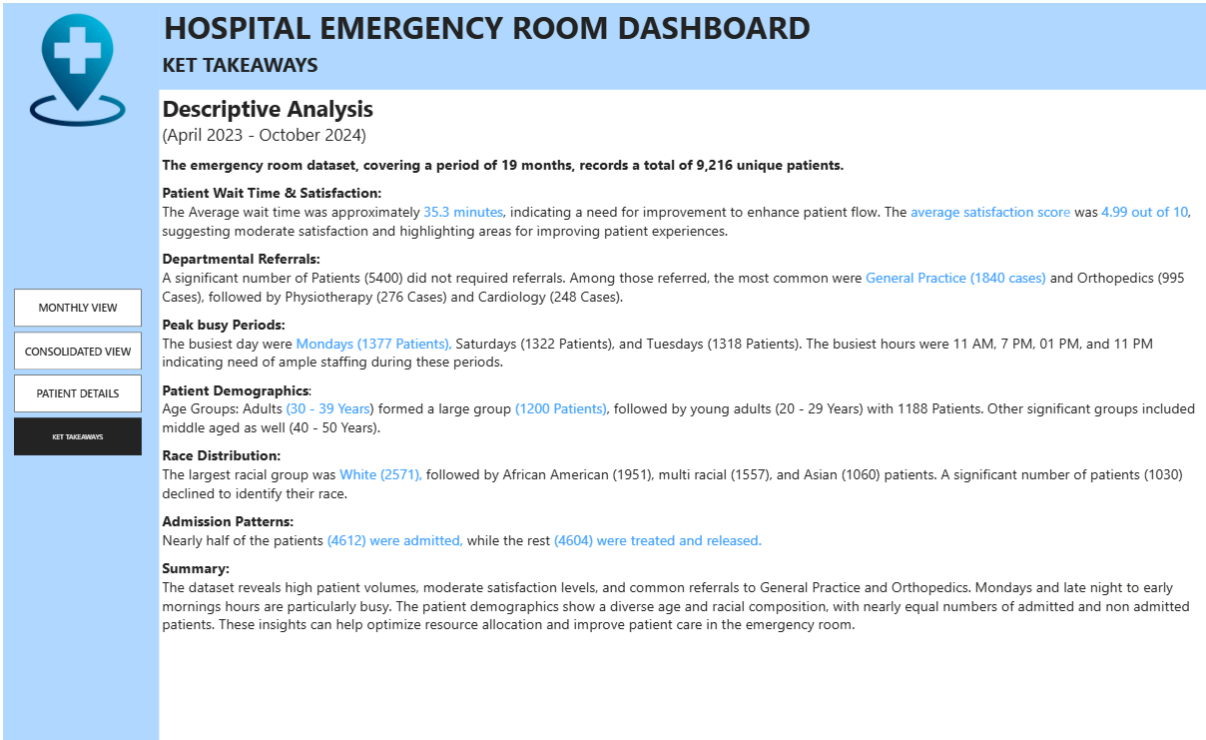
Display detailed patient-level information for operational monitoring and record analysis.

Features

- Search and filter patient records.
- View complete patient details.
- Analyze individual cases quickly.

The **Patient Details Dashboard** provides detailed information for every patient who visited the Emergency Room. It displays patient-specific records, including demographic details, admission status, department referrals, and wait times. This dashboard allows users to search, filter, and review individual patient information, making it useful for operational monitoring, record verification, and detailed case analysis.

Dashboard 4: Key Takeaways



Objective

Present a summary of important findings and actionable insights from the dashboard.

Key Insights

- Peak patient arrival periods.
- Departments with the highest referrals.
- Average wait time performance.
- Patient satisfaction trends.
- Overall Emergency Room efficiency.

The **Key Takeaways Dashboard** summarizes the most important findings from the entire analysis. It highlights overall hospital performance, key trends, operational challenges, and business insights derived from the data. The dashboard also presents actionable recommendations that can help improve patient care, optimize resource allocation, reduce wait times, and enhance the overall efficiency of Emergency Room operations.

10. Dashboard Analysis

The dashboard provides a comprehensive analysis of Emergency Room operations by monitoring patient admissions, wait times, satisfaction scores, department referrals, and patient demographics. It also identifies peak patient arrival times, evaluates service efficiency, and helps hospital management understand operational performance through interactive visualizations.

11. Key Insights

- Patient visits vary across different days and months.
- Peak hours result in longer wait times.
- Cardiology and Orthopedics receive the highest referrals.
- Longer wait times can impact patient satisfaction.
- Most patients are attended to within the target response time.
- Demographic analysis supports better healthcare planning.

12. Business Recommendations

- Increase staff during busy hours.
- Reduce patient wait times.
- Monitor high-demand departments.
- Improve patient satisfaction through efficient service.
- Use dashboard insights for better resource planning and decision-making.

13. Conclusion

The Hospital Emergency Room Dashboard transforms raw patient data into meaningful insights using Power BI. It helps hospital management monitor performance, improve patient care, optimize resources, and support data-driven decision-making through interactive and easy-to-understand dashboards.

14. Skills Demonstrated

Power BI

- Dashboard Development
- Interactive Reports
- KPI Monitoring
- Data Visualization

Power Query

- Data Cleaning
- Data Transformation

DAX

- Measures
- KPI Calculations

Data Modeling

- Star Schema
- Table Relationships

Business Intelligence

- Healthcare Analytics
- Trend Analysis
- Performance Reporting
- Data-Driven Decision Making